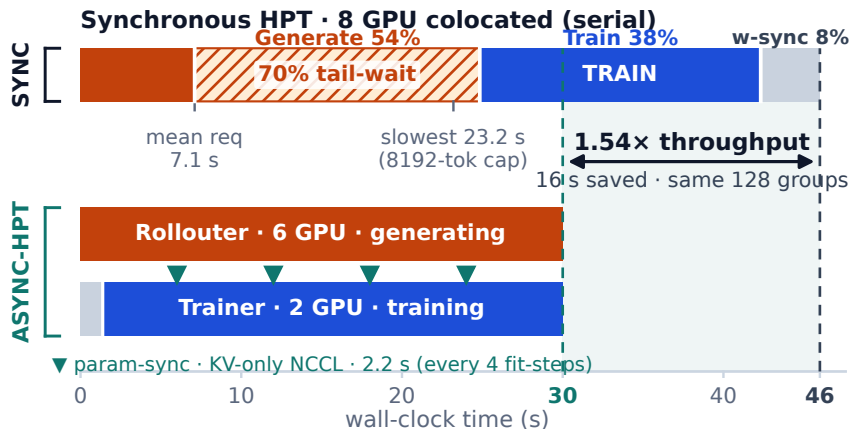
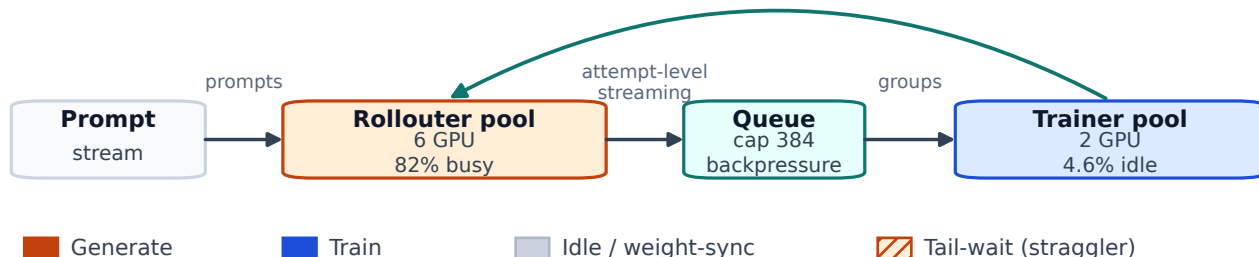


Execution efficiency: Synchronous HPT vs. Async-HPT (8×B200)

Same unit of work — 128 prompt-groups — on a shared wall-clock scale

param-sync loop-back · every 4 fit-steps · KV-cache-only NCCL 2.2 s · partial-rollout preserves prefixes



SYNC → ASYNC

Throughput (grp/s) **1.54×**

2.79 → 4.31

Trainer idle **-91%**

54% → 4.6%

Actor MFU **1.57×**

13.3% → 20.9%

weight-sync 3.4 s/step (sync)
param-sync 2.2 s (async)

Advantage widens 1.31× (10 min) → 1.59× (40 min) as the sync straggler tax grows with response length (1972 → 3042 tok).